Analytics & Big Data

Session 9: Big data II

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https://jose.theoj.org/papers/10.21105/jose.00042

🎯 **Learning objectives**

Tentatively

- **Explain** the characteristics and challenges of unstructured data sources (e.g., text, social media, spatial data).
- **Acquire and preprocess** large-scale or unstructured data using APIs and appropriate tools in Python.
- **Apply** large language models or other scalable methods to extract, classify, or summarize unstructured data.
- **Evaluate** the reliability, bias, and limitations of API- and LLM-based analytics workflows.

TBD: spatial/social-media analytics?

TODO

TBD: broad group work and presentations? (-> TBD: how to make that relevant/documented for the exam??)

Reinforcement Learning

The solution to many of the problems in our lives cannot be automated. This is not because current computers are too slow, but simply because it is too difficult for humans to determine what the program should do.

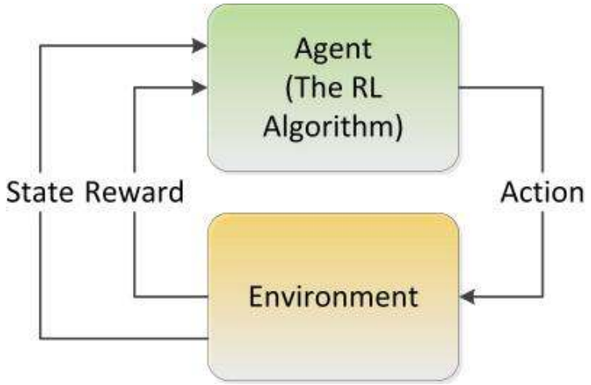
Supervised learning is a general method for training an approximator. However, supervised learning requires sample input-output pairs from the domain to be learned.

For example, we might not know the best way to program a computer to recognize an infrared picture of a tank, but we do have a large collection of infrared pictures, and we do know whether each picture contains a tank or not. Supervised learning could look at all the examples with answers, and learn how to recognize tanks in general.

Unfortunately, there are many situations where we don’t know the correct answers that supervised learning requires. For example, in a self-driving car, the question would be the set of all sensor readings at a given time, and the answer would be how the controls should react during the next millisecond.

For these cases there exist a different approach known as reinforcement learning.

Reinforcement Learning



> The agent learns how to achieve a given goal by trial-and-error interactions with its environment by maximizing a reward.

AlphaGo

Go is one of the hardest games in the world for AI because of the huge number of different game scenarios and moves. The number of potential legal board positions is greater than the number of atoms in the universe.

The core of AlphaGo is a deep neural network. It was initially trained to learn playing by using a database of around 30 million recorded historical moves. After the training, the system was cloned and it was trained further playing large numbers of games against other instances of itself, using reinforcement learning to improve its play. During this training AlphaGo learned new strategies which were never played by humans.

A newer version named AlphaGo Zero skips the step of being trained and learns to play simply by playing games against itself, starting from completely random play.

Libratus

An artificial intelligence called Libratus has beaten four of the world's best poker players in a grueling 20-day tournament in January 2017.

Poker is more difficult because it's a game with imperfect information. With chess and Go, each player can see the entire board, but with poker, players don't get to see each other's hands. Furthermore, the AI is required to bluff and correctly interpret misleading information in order to win.

"We didn't tell Libratus how to play poker. We gave it the rules of poker and said 'learn on your own'." The AI started playing randomly but over the course of playing trillions of hands was able to refine its approach and arrive at a winning strategy.

Types of Artificial Intelligence

Discriminative AI is designed to differentiate and classify input, but not to create new content. Examples include image or speech recognition, credit scoring or stock price prediction.

Generative AI is able to generate new content based on existing information and user specifications. This includes texts, images, videos, program code, etc. The generated content can often hardly be distinguished from human-generated content. As things stand at present, however, they are pure recombinations of learned knowledge.

Well-known examples of generative AI are language models for generating text, such as GPT-3 or GPT-4, and the chatbot ChatGPT based on them, or image generators such as Stable Diffusion and DALL-E.

ChatGPT

ChatGPT is a generative AI that produces human-like text and communicates with humans.

The “GPT” in ChatGPT comes from the language model of the same name, which was extended for ChatGPT with various components for communication and quality assurance.

GPT is based on a huge neural network that essentially represents the language model. While the first GPT-3 has 175 billion parameters, the newer GPT-4 already has 1 trillion parameters. Compared to GPT-3, GPT-4 is therefore more intelligent, can deal with more extensive questions and conversations and makes fewer factual errors.

ChatGPT - Approach

ChatGPT generates its response word by word via a sequence of probabilities, with each new word depending on the previous ones.

The best thing about AI is its ability to

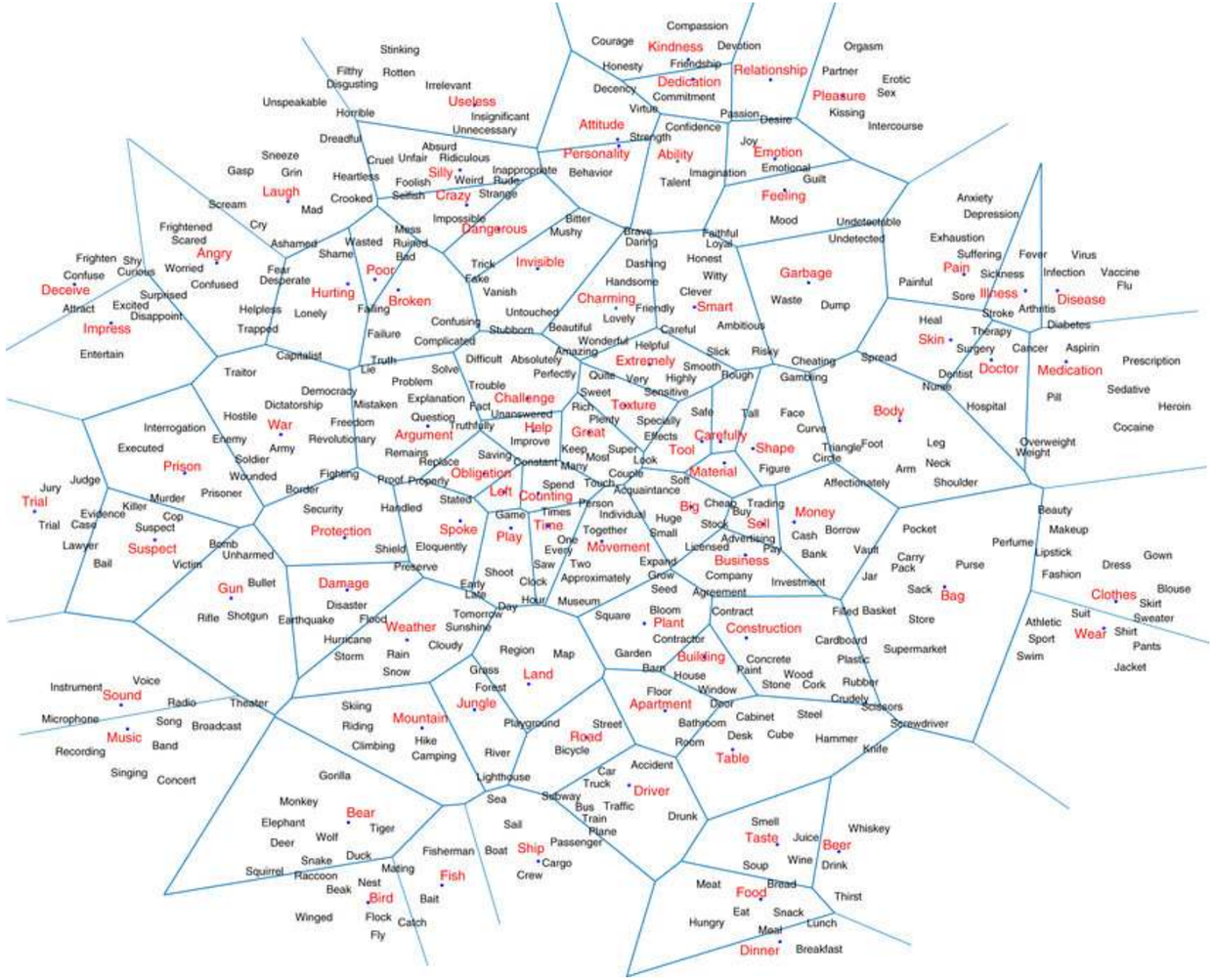
learn	4.5%
predict	3.5%
make	3.2%
understand	3.1%
do	2.9%

The best thing about AI is its ability to,
The best thing about AI is its ability to learn,
The best thing about AI is its ability to learn from,
The best thing about AI is its ability to learn from experience,
The best thing about AI is its ability to learn from experience.,
The best thing about AI is its ability to learn from experience. It,
The best thing about AI is its ability to learn from experience. It's,
The best thing about AI is its ability to learn from experience. It's not

The most probable word is not always selected; instead, randomization takes place. This means that different variants can be created for the same task.

The best thing about AI is its ability to learn. I've always liked the
The best thing about AI is its ability to really come into your world and just
The best thing about AI is its ability to examine human behavior and the way it
The best thing about AI is its ability to do a great job of teaching us
The best thing about AI is its ability to create real tasks, but you can

ChatGPT - Semantic Spaces (I)


Compassion
Courage
Kindness
Devotion
Orgasm
Honesty
Friendship
Relationship
Partner
Erotic
Decency
Dedication
Commitment
Pleasure
Sex
Virtue
Passion
Desire
Kissing
Intercourse
Attitude
Confidence
Strength
Joy
Personality
Ability
Emotion
Talent
Imagination
Emotional
Guilt
Feeling
Mood
Undetectable
Undetected
Anxiety
Depression
Exhaustion
Suffering
Fever
Virus
Pain
Sickness
Infection
Vaccine
Flu
Illness
Disease
Stroke
Arthritis
Diabetes
Heal
Therapy
Skin
Surgery
Cancer
Aspirin
Doctor
Medication
Prescription
Dentist
Nurse
Hospital
Pill
Sedative
Heroin
Cocaine
Overweight
Weight
Body
Leg
Neck
Shoulder
Arm
Foot
Triangle
Circle
Affectionately
Figure
Face
Curve
Tall
Safe
Shape
Tool
Carefully
Material
Texture
Great
Help
Improve
Keep
Super
Most
Look
Effects
Specially
Plenty
Rich
Sweet
Sensitive
Quite
Very
Highly
Rough
Gambling
Cheating
Spread
Smooth
Extremely
Perfectly
Difficult
Absolutely
Amazing
Wonderful
Helpful
Careful
Ambitious
Slick
Risky
Beautiful
Stubborn
Complicated
Confusing
Untouched
Charming
Friendly
Smart
Lovely
Clever
Handsome
Witty
Waste
Dump
Garbage
Faithful
Loyal
Brave
Daring
Dashing
Honest
Bitter
Mushy
Dangerous
Invisible
Trick
Fake
Vanish
Poor
Broken
Failing
Failure
Truth
Lie
Solve
Problem
Trouble
Challenge
Fact
Unanswered
Truthfully
Question
Explanation
Mistaken
Freedom
Revolutionary
Argument
Remains
Replace
Saving
Constant
Many
Obligation
Proof
Property
Fighting
Border
Stated
Handled
Security
Protection
Shield
Eloquently
Preserve
Spoke
Play
Time
Movement
Loft
Counting
Spend
Touch
Couple
Acquaintance
Person
Individual
Together
One
Every
Saw
Two
Approximately
Shoot
Clock
Early
Late
Hour
Museum
Day
Tomorrow
Sunshine
Weather
Flood
Earthquake
Disaster
Hurricane
Storm
Rain
Snow
Cloudy
Damage
Gun
Bullet
Rifle
Shotgun
Bomb
Unharmed
Victim
Suspect
Prison
Prisoner
Murder
Killer
Evidence
Cop
Case
Trial
Judge
Jury
Lawyer
Bail
Executed
Interrogation
Hostile
War
Enemy
Army
Soldier
Wounded
Dictatorship
Democracy
Traitor
Capitalist
Entertain
Impress
Deceive
Attract
Excited
Disappoint
Surprised
Confused
Worried
Shy
Frighten
Confuse
Curious
Scared
Frightened
Angry
Cry
Scream
Mad
Laugh
Crooked
Gasp
Grin
Sneeze
Dreadful
Horrible
Unspeakable
Filthy
Disgusting
Stinking
Rotten
Irrelevant
Useless
Insignificant
Unnecessary
Absurd
Unfair
Ridiculous
Cruel
Silly
Heartless
Foolish
Weird
Rude
Inappropriate
Behavior
Crazy
Selfish
Strange
Impossible
Mess
Ruined
Bad
Ashamed
Shame
Wasted
Fear
Desperate
Hurting
Helpless
Lonely
Trapped
Sound
Instrument
Voice
Radio
Theater
Microphone
Music
Song
Broadcast
Recording
Band
Singing
Concert
Gorilla
Monkey
Elephant
Bear
Tiger
Deer
Wolf
Snake
Duck
Squirrel
Raccoon
Beak
Nest
Bird
Winged
Flock
Fly
Catch
Bait
Fish
Mating
Fisherman
Boat
Ship
Passenger
Cargo
Crew
Plane
Train
Bus
Traffic
Subway
Sea
Sail
Lighthouse
Car
Truck
Driver
Accident
Bicycle
Road
River
Camping
Hike
Climbing
Riding
Skiing
Mountain
Playground
Jungle
Street
Forest
Grass
Land
Region
Map
Garden
Barn
Building
House
Plant
Square
Bloom
Contractor
Construction
Contract
Cardboard
Supermarket
Store
Filled Basket
Money
Cash
Borrow
Pocket
Bank
Vault
Carry
Pack
Jar
Sack
Bag
Purse
Perfume
Beauty
Makeup
Lipstick
Fashion
Dress
Gown
Clothes
Blouse
Skirt
Sweater
Suit
Athletic
Sport
Swim
Wear
Shirt
Pants
Jacket
Big
Cheap
Trading
Buy
Soil
Advertising
Licensed
Business
Company
Pay
Investment
Agreement
Seed
Grow
Expand
Approximately
Two
Saw
Every
One
Time
Play
Spoke
Protection
Shield
Eloquently
Preserve
Obligation
Loft
Counting
Spend
Touch
Couple
Acquaintance
Person
Individual
Together
Huge
Small
Stock
Cheap
Trading
Buy
Soil
Advertising
Licensed
Business
Company
Pay
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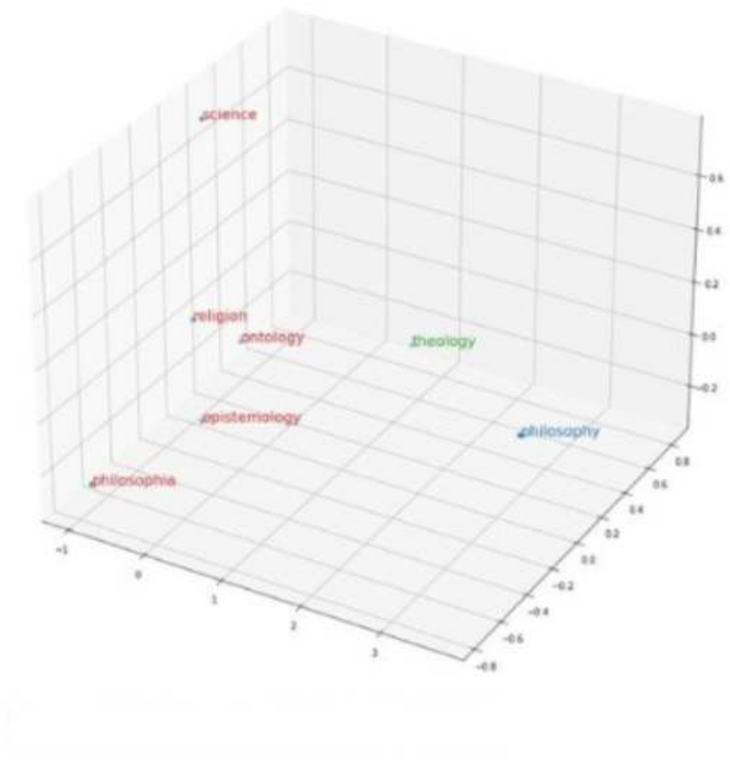
One
Time
Play
Spoke
Protection
Shield
Eloquently
Preserve

ChatGPT - Semantic Spaces (II)

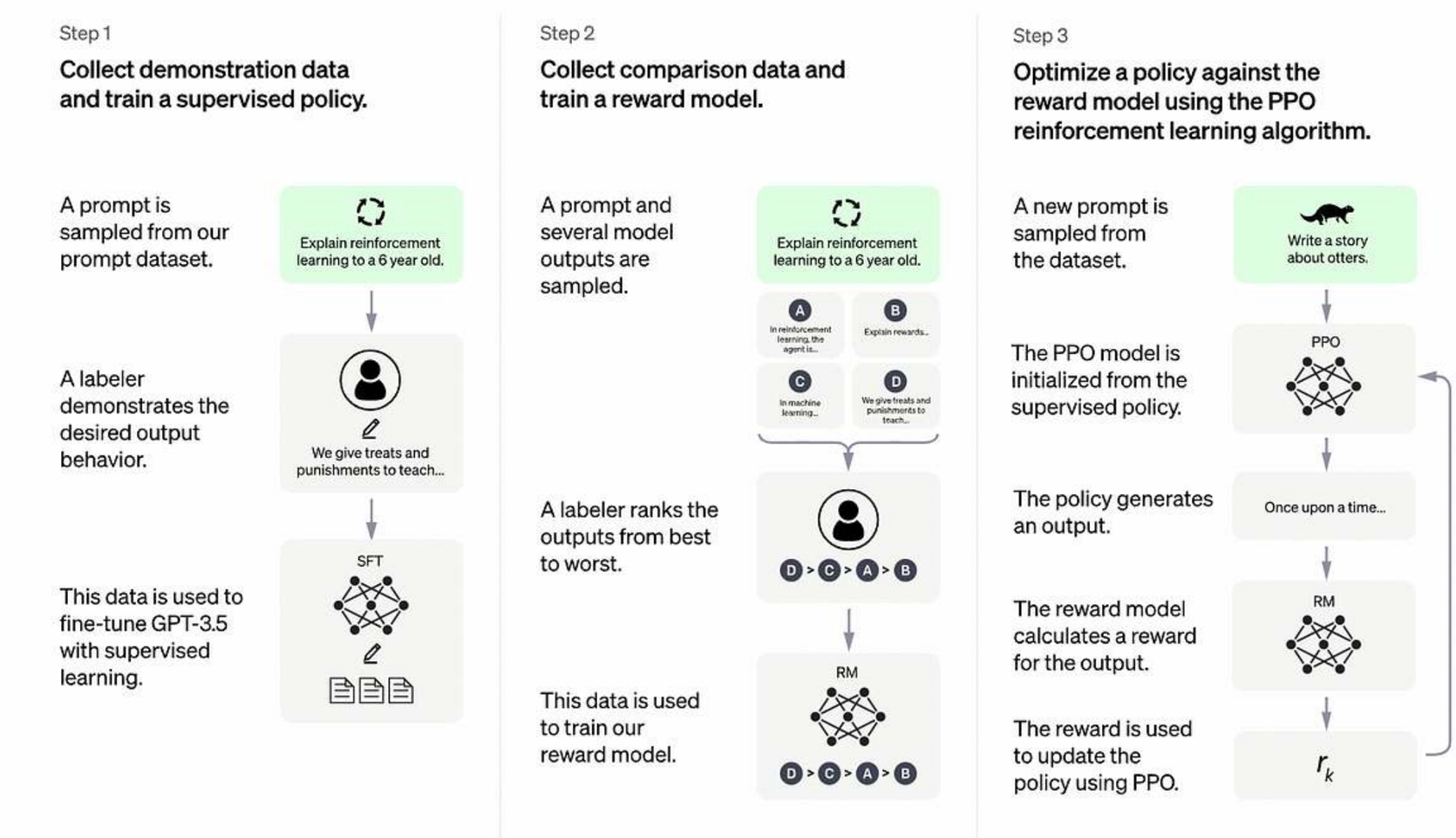



```
print_neighbors('ontology', model5)
```

Approximate Neighbors

```
('Eros', 0.8966988027095795)
('Astronomy', 0.8926706984639168)
('Philosophy', 0.886092908680439)
('mathematics', 0.8845213651657104)
('Psychology', 0.8803791999816895)
('Night', 0.8794359490275383)
('subject-object', 0.8780221417546272)
('king', 0.8750540241599083)
('epistemology', 0.8732230812311172)
('Dionysus', 0.8667336702346802)
```

ChatGPT - Evaluation Component



Survey: Session 9

https://forms.gle/1vsCsqc3SzWfSX1f6

Note: Responses may be analyzed and published in anonymized form.

Please complete the survey before you leave today — thank you 🙏

References